

International Space Station

“A space station will permit quantum leaps in our research in science, communications, and in metals and lifesaving medicines...”

PRESIDENT RONALD REAGAN
STATE OF THE UNION SPEECH, 1984

THE ASSEMBLY OF THE International Space Station (ISS) began on November 20, 1998, when a Russian Proton rocket, launched from Baikonur in Kazakhstan, lifted the Zarya module into orbit. A crew of three took up residence on the ISS in November 2000, and the station has been in continuous occupation ever since – by far the most sustained human presence in space. The basis for the project lay in international cooperation, primarily between the United States and Russia, with Japan, Canada, and members of the European Space Agency also involved.

The ISS has continuously expanded since 2000. Modules added include science laboratories, where experiments are performed exploiting the micro-gravity environment. The effects of prolonged weightlessness on the human body are a particular object of study, with a bearing upon possible future human missions to the Moon and Mars, for which the ISS might provide a staging post. The retirement of the Space Shuttle in 2011 left the Russian Soyuz as the only human spacecraft supporting the ISS. A number of automated vehicles, notably the SpaceX Dragon, ferry cargo to the space station. Since 2001 the ISS has also seen the beginnings of space tourism with a few wealthy individuals paying large sums to travel to the ISS aboard Soyuz craft.

ASSEMBLY MISSION

NASA Space Shuttles made a total of 37 flights to the ISS, playing an essential role in assembly of the station. The final Shuttle flight to the ISS took place in July 2011.

IN PROGRESS

The International Space Station is shown here in the early stages of its assembly. Its flexible modular design allowed new elements to be added to the structure over many years.

